TRANSLATION BY A POINT WITH BIJECTIVE SOURCE AND TARGET IS AN AUTOMORPHISM OF THE ARROW SCHEME

THE STACKS PROJECT AUTHORS

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Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(k)$ with k a field. If $r \in R$ is a point such that s, t induce isomorphisms $k \to \kappa(r)$, then the map*

$$R \longrightarrow R, \quad x \longmapsto c(r, x)$$

(see proof for precise notation) is an automorphism $R \to R$ which maps e to r.

Proof. This is completely obvious if you think about groupoids in a functorial way. But we will also spell it out completely. Denote $a : U \to R$ the morphism with image r such that $s \circ a = \mathrm{id}_U$ which exists by the hypothesis that $s : k \to \kappa(r)$ is an isomorphism. Similarly, denote $b : U \to R$ the morphism with image r such that $t \circ b = \mathrm{id}_U$. Note that $b = a \circ (t \circ a)^{-1}$, in particular $a \circ s \circ b = b$.

Consider the morphism $\Psi : R \to R$ given on T-valued points by

$$(f : T \to R) \longmapsto (c(a \circ t \circ f, f) : T \to R)$$

To see this is defined we have to check that $s \circ a \circ t \circ f = t \circ f$ which is obvious as $s \circ a = 1$. Note that $\Phi(e) = a$, so that in order to prove the lemma it suffices to show that Φ is an automorphism of R. Let $\Phi : R \to R$ be the morphism given on T-valued points by

$$(g : T \to R) \longmapsto (c(i \circ b \circ t \circ g, g) : T \to R).$$

This is defined because $s \circ i \circ b \circ t \circ g = t \circ b \circ t \circ g = t \circ g$. We claim that Φ and Ψ are inverse to each other. To see this we compute

$$\begin{aligned}
& c(a \circ t \circ c(i \circ b \circ t \circ g, g), c(i \circ b \circ t \circ g, g)) \\
&= c(a \circ t \circ i \circ b \circ t \circ g, c(i \circ b \circ t \circ g, g)) \\
&= c(a \circ s \circ b \circ t \circ g, c(i \circ b \circ t \circ g, g)) \\
&= c(b \circ t \circ g, c(i \circ b \circ t \circ g, g)) \\
&= c(c(b \circ t \circ g, i \circ b \circ t \circ g), g)) \\
&= c(e, g) \\
&= g
\end{aligned}$$

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where we have used the relation $a \circ s \circ b = b$ shown above. In the other direction we have

$$\begin{aligned}
& c(i \circ b \circ t \circ c(a \circ t \circ f, f), c(a \circ t \circ f, f)) \\
& = c(i \circ b \circ t \circ a \circ t \circ f, c(a \circ t \circ f, f)) \\
& = c(i \circ a \circ (t \circ a)^{-1} \circ t \circ a \circ t \circ f, c(a \circ t \circ f, f)) \\
& = c(i \circ a \circ t \circ f, c(a \circ t \circ f, f)) \\
& = c(c(i \circ a \circ t \circ f, a \circ t \circ f), f) \\
& = c(e, f) \\
& = f
\end{aligned}$$

The lemma is proved. □